

music1

February 25, 2025

1 Set Up

```
[1]: # Import
import pandas as pd
import matplotlib.pyplot as plt
import numpy as np
```

```
[2]: # Read Files

# Path to your Parquet file
parquet_file = '/Users/henryding/Documents/Seminar/Data/output_data1.parquet'

# Read the Parquet file into a DataFrame
df = pd.read_parquet(parquet_file, engine='pyarrow') # You can also use ↪
↪ 'fastparquet'

# Display the first few rows of the DataFrame
display(df.head(1000))
num_rows = df.shape[0]
print(f"The number of rows in the DataFrame is: {num_rows}")
```

	status	gender	length	firstName	level	lastName	registration \
0	200	M	524.32934	Shlok	paid	Johnson	1.533735e+12
1	200	F	238.39302	Vianney	paid	Miller	1.537500e+12
2	200	F	140.35546	Vina	paid	Bailey	1.536415e+12
3	200	M	277.15873	Andres	paid	Foley	1.534387e+12
4	200	F	1121.25342	Aaliyah	paid	Ramirez	1.537381e+12
..	...	...	...	...	...	...	...
995	307	F	NaN	Alivia	paid	Williams	1.535955e+12
996	200	M	315.48036	Christian	free	Klein	1.536940e+12
997	200	M	248.73751	Kristofer	free	James	1.536879e+12
998	200	F	53.08036	Zoey	free	Gregory	1.533190e+12
999	200	F	NaN	Zoey	free	Gregory	1.533190e+12

	userId	ts	auth	page	sessionId \
0	1749042	1538352001000	Logged In	NextSong	22683
1	1563081	1538352002000	Logged In	NextSong	20836
2	1697168	1538352002000	Logged In	NextSong	4593

3	1222580	1538352003000	Logged In	NextSong	6370
4	1714398	1538352003000	Logged In	NextSong	22316
..	...	...	...	...	...
995	1792538	1538352525000	Logged In	Thumbs Up	8871
996	1291813	1538352526000	Logged In	NextSong	14821
997	1929921	1538352526000	Logged In	NextSong	9831
998	1839740	1538352527000	Logged In	NextSong	24848
999	1839740	1538352527000	Logged In	Roll Advert	24848

	location	itemInSession	\
0	Dallas-Fort Worth-Arlington, TX	278	
1	San Francisco-Oakland-Hayward, CA	9	
2	Hilo, HI	109	
3	Watertown, SD	71	
4	Baltimore-Columbia-Towson, MD	21	
..	...	...	
995	New York-Newark-Jersey City, NY-NJ-PA	75	
996	New York-Newark-Jersey City, NY-NJ-PA	10	
997	Seattle-Tacoma-Bellevue, WA	56	
998	Phoenix-Mesa-Scottsdale, AZ	11	
999	Phoenix-Mesa-Scottsdale, AZ	12	

	userAgent	\
0	"Mozilla/5.0 (Windows NT 6.1; WOW64) AppleWebKit...	
1	"Mozilla/5.0 (Macintosh; Intel Mac OS X 10_9_4...	
2	Mozilla/5.0 (Macintosh; Intel Mac OS X 10.9; r...	
3	"Mozilla/5.0 (Macintosh; Intel Mac OS X 10_9_4...	
4	"Mozilla/5.0 (Windows NT 6.1; WOW64) AppleWebKit...	
..	...	
995	Mozilla/5.0 (Macintosh; Intel Mac OS X 10.9; r...	
996	"Mozilla/5.0 (Macintosh; Intel Mac OS X 10_9_2...	
997	Mozilla/5.0 (X11; Ubuntu; Linux x86_64; rv:31...	
998	"Mozilla/5.0 (X11; Linux x86_64) AppleWebKit/5...	
999	"Mozilla/5.0 (X11; Linux x86_64) AppleWebKit/5...	

	song	\
0	Ich mache einen Spiegel - Dream Part 4	
1	MiÃnte	
2	Baby Talk	
3	Horn Concerto No. 4 in E flat K495: II. Romanc...	
4	Close To The Edge (I. The Solid Time Of Change...	
..	...	
995	None	
996	Odessa	
997	Jigsaw Falling Into Place	
998	Broken Hearted Hoover Fixer Sucker Guy	
999	None	

	artist	method
0	Popol Vuh	PUT
1	Los Bunkers	PUT
2	Lush	PUT
3	Barry Tuckwell/Academy of St Martin-in-the-Fie...	PUT
4	Yes	PUT
..	...	...
995	None	PUT
996	Caribou	PUT
997	Radiohead	PUT
998	Glen Hansard	PUT
999	None	GET

[1000 rows x 18 columns]

The number of rows in the DataFrame is: 26259199

2 Exploratory Analysis

[3]: *# Categorical Variables*

```
unique_level = df['level'].unique()
print(f"level: {unique_level}")

unique_auth = df['auth'].unique()
print(f"auth: {unique_auth}")

unique_method = df['method'].unique()
print(f"method: {unique_method}")

unique_page = df['page'].unique()
print(f"page: {unique_page}")
```

```
level: ['paid' 'free']
auth: ['Logged In' 'Logged Out' 'Cancelled' 'Guest']
method: ['PUT' 'GET']
page: ['NextSong' 'Home' 'Thumbs Up' 'Error' 'Help' 'Add to Playlist'
'Roll Advert' 'Login' 'Thumbs Down' 'Settings' 'About' 'Add Friend'
'Upgrade' 'Submit Upgrade' 'Logout' 'Downgrade' 'Save Settings'
'Submit Downgrade' 'Cancel' 'Cancellation Confirmation' 'Register'
'Submit Registration']
```

[4]: *# User Usage*

```
user_counts = df.groupby('userId').size().reset_index(name='count')
display(user_counts.sort_values(by='count', ascending=False))
```

	userId	count
5936	1261737	778479

```

12534  1564221    13591
20802  1931933    12831
163    1006695    12372
7562   1336969    11858
...
6072   1267517     1
9698   1434698     1
17778  1793623     1
9129   1408726     1
15322  1689121     1

```

[22278 rows x 2 columns]

```

[39]: # Time Period
# Convert Timestamps to datetime
df['ts'] = pd.to_datetime(df['ts'], unit='ms')

print(f"timestamp min: {df['ts'].min()}")
print(f"timestamp max: {df['ts'].max()}")

# Convert Registration to datetime
df['registration'] = pd.to_datetime(df['registration'], unit='ms')

print(f"registration min: {df['registration'].min()}")
print(f"registration max: {df['registration'].max()}")

```

```

timestamp min: 2018-10-01 00:00:01
timestamp max: 2018-12-01 00:00:02
registration min: 2017-10-14 22:05:25
registration max: 2018-12-03 07:23:42

```

```

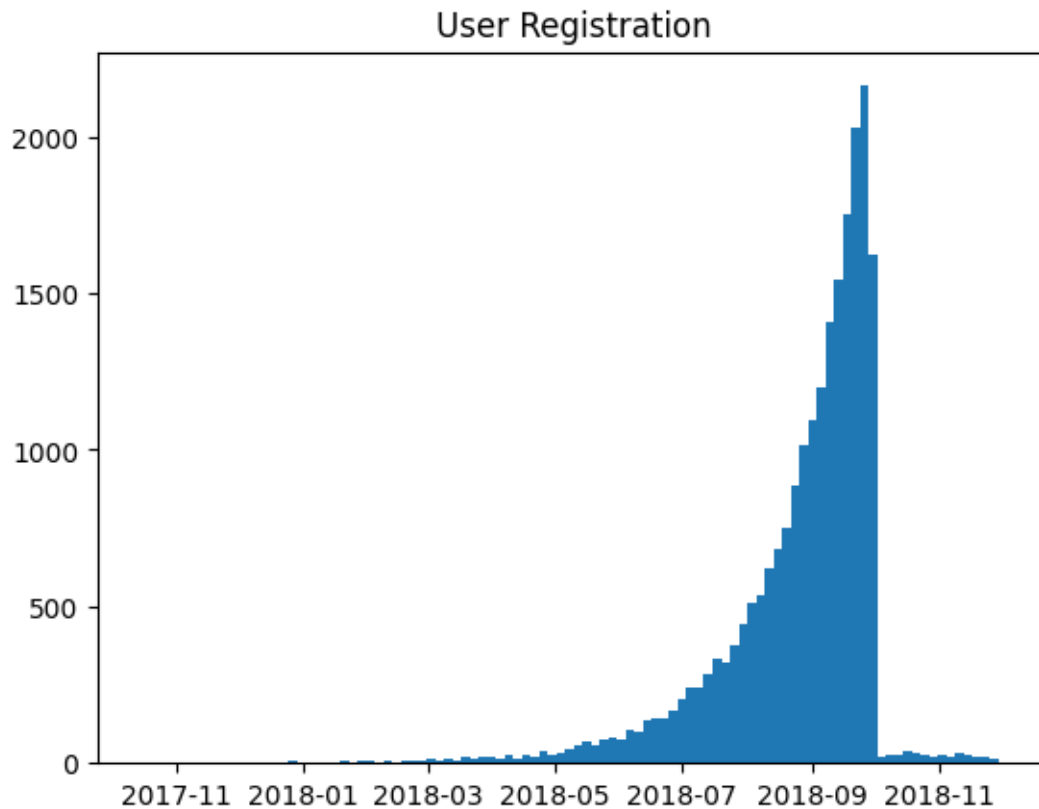
[47]: # User Registration
grouped_registration = df.groupby('userId')['registration'].agg('min')
display(grouped_registration.head())
plt.hist(grouped_registration, bins=100)
plt.title('User Registration')
plt.show()

```

```

userId
1000025    2018-07-10 09:30:08
1000035    2018-09-12 19:28:22
1000083    2018-09-07 18:01:49
1000103    2018-09-22 07:27:25
1000164    2018-08-12 09:32:01
Name: registration, dtype: datetime64[ns]

```



```
[6]: # High-Cardinality Variables
song_count = df.groupby('song').size().count()
print(f"Number of Songs: {song_count}")

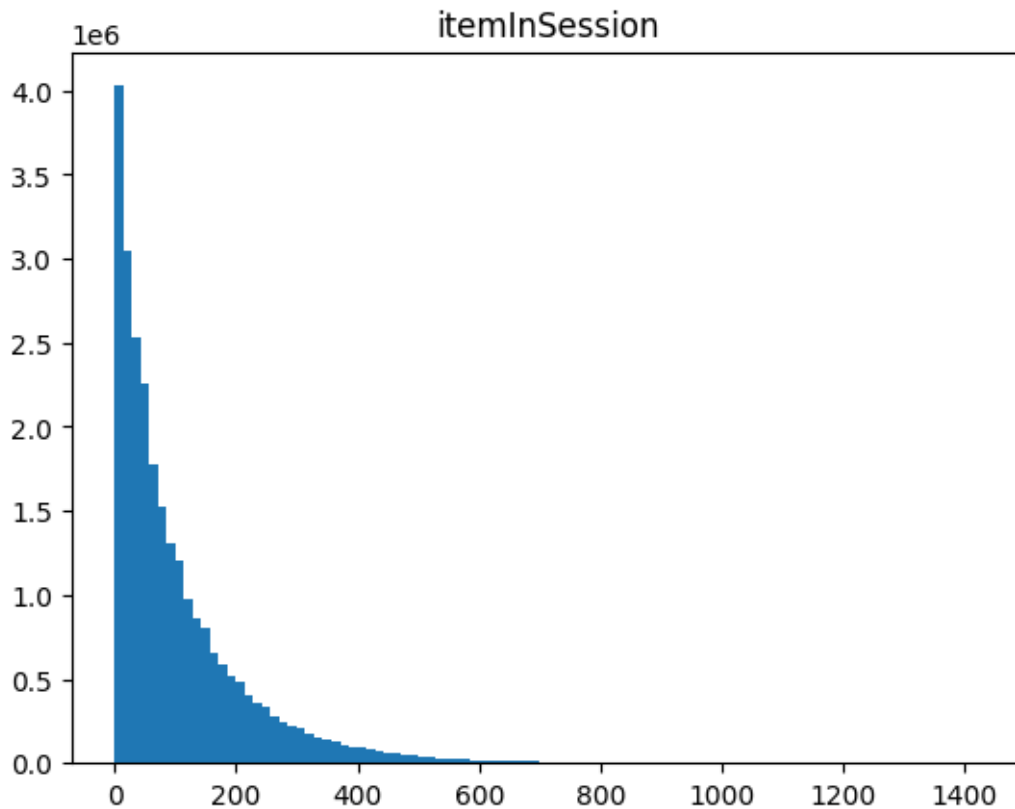
artist_count = df.groupby('artist').size().count()
print(f"Number of Artists: {artist_count}")

user_count = df.groupby('userId').size().count()
print(f"Number of Users: {user_count}")
```

Number of Songs: 253564
Number of Artists: 38337
Number of Users: 22278

```
[11]: # Continuous Variables

plt.hist(df['itemInSession'], bins=100)
plt.title("itemInSession")
plt.show()
```



3 Data Preprocessing

[35]:

4 Feature Engineering

```
[ ]: # Page
for click_type in unique_page:
    df[click_type] = np.where(df['page'] == click_type, 1, 0)

display(df)
```

```
[36]: # group by user_id

# grouped = df.groupby(['userId', 'gender']).agg({
#     ''
# })
```

	status	gender	length	firstName	level	lastName	registration \
0	200	M	524.32934	Shlok	paid	Johnson	1.533735e+12
1	200	F	238.39302	Vianney	paid	Miller	1.537500e+12
2	200	F	140.35546	Vina	paid	Bailey	1.536415e+12
3	200	M	277.15873	Andres	paid	Foley	1.534387e+12
4	200	F	1121.25342	Aaliyah	paid	Ramirez	1.537381e+12
...	...	...	...	...	...	...	...
26259194	200	F	143.38567	Macee	paid	Logan	1.537589e+12
26259195	200	M	234.70975	Collin	paid	Deleon	1.535986e+12
26259196	200	M	224.57424	Michael	paid	Shaw	1.532106e+12
26259197	200	F	320.83546	Peyton	paid	Jones	1.534957e+12
26259198	200	M	182.69995	Dexter	paid	Lynch	1.533090e+12

	userId	ts	auth	page	sessionId \
0	1749042	2018-10-01 00:00:01	Logged In	NextSong	22683
1	1563081	2018-10-01 00:00:02	Logged In	NextSong	20836
2	1697168	2018-10-01 00:00:02	Logged In	NextSong	4593
3	1222580	2018-10-01 00:00:03	Logged In	NextSong	6370
4	1714398	2018-10-01 00:00:03	Logged In	NextSong	22316
...	...	...	...	...	...
26259194	1521799	2018-11-30 23:59:50	Logged In	NextSong	44039
26259195	1368090	2018-11-30 23:59:54	Logged In	NextSong	43964
26259196	1979129	2018-11-30 23:59:56	Logged In	NextSong	44533
26259197	1068798	2018-11-30 23:59:56	Logged In	NextSong	44408
26259198	1623507	2018-11-30 23:59:59	Logged In	NextSong	42056

	location	itemInSession \
0	Dallas-Fort Worth-Arlington, TX	278
1	San Francisco-Oakland-Hayward, CA	9
2	Hilo, HI	109
3	Watertown, SD	71
4	Baltimore-Columbia-Towson, MD	21
...	...	...
26259194	Detroit-Warren-Dearborn, MI	89
26259195	San Diego-Carlsbad, CA	125
26259196	Detroit-Warren-Dearborn, MI	76
26259197	Miami-Fort Lauderdale-West Palm Beach, FL	160
26259198	Huntsville, AL	152

	userAgent \
0	"Mozilla/5.0 (Windows NT 6.1; WOW64) AppleWebKit...
1	"Mozilla/5.0 (Macintosh; Intel Mac OS X 10_9_4...
2	Mozilla/5.0 (Macintosh; Intel Mac OS X 10.9; r...
3	"Mozilla/5.0 (Macintosh; Intel Mac OS X 10_9_4...
4	"Mozilla/5.0 (Windows NT 6.1; WOW64) AppleWebKit...
...	...
26259194	"Mozilla/5.0 (Windows NT 6.1; WOW64) AppleWebKit...
26259195	"Mozilla/5.0 (Windows NT 6.3; WOW64) AppleWebKit...

26259196 "Mozilla/5.0 (Macintosh; Intel Mac OS X 10_9_4...
 26259197 Mozilla/5.0 (Windows NT 6.1; WOW64; rv:31.0) G...
 26259198 "Mozilla/5.0 (X11; Linux x86_64) AppleWebKit/5...

song \
 0 Ich mache einen Spiegel - Dream Part 4
 1 MiÃntele
 2 Baby Talk
 3 Horn Concerto No. 4 in E flat K495: II. Romanc...
 4 Close To The Edge (I. The Solid Time Of Change...
 ...
 26259194 Indigo Flow
 26259195 More And More
 26259196 You Told A Lie
 26259197 Hailie's Song
 26259198 The Way You Do The Things You Do (2009 Digital...

	artist	method	NextSong	\
0	Popol Vuh	PUT	1	
1	Los Bunkers	PUT	1	
2	Lush	PUT	1	
3	Barry Tuckwell/Academy of St Martin-in-the-Fie...	PUT	1	
4	Yes	PUT	1	
...	...	...	...	
26259194	Limp Bizkit	PUT	1	
26259195	Captain Hollywood Project	PUT	1	
26259196	Camera Obscura	PUT	1	
26259197	Eminem	PUT	1	
26259198	UB40	PUT	1	

	Home	Thumbs Up	Error	Help	Add to Playlist	Roll	Advert	Login	\
0	0	0	0	0	0		0	0	
1	0	0	0	0	0		0	0	
2	0	0	0	0	0		0	0	
3	0	0	0	0	0		0	0	
4	0	0	0	0	0		0	0	
...	...	...	...		...	...			
26259194	0	0	0	0	0		0	0	
26259195	0	0	0	0	0		0	0	
26259196	0	0	0	0	0		0	0	
26259197	0	0	0	0	0		0	0	
26259198	0	0	0	0	0		0	0	

	Thumbs Down	Settings	About	Add Friend	Upgrade	Submit	Upgrade	\
0	0	0	0	0	0		0	
1	0	0	0	0	0		0	
2	0	0	0	0	0		0	
3	0	0	0	0	0		0	

4		0		0		0		0		0
...	...		...	...		...		...		
26259194		0		0		0		0		0
26259195		0		0		0		0		0
26259196		0		0		0		0		0
26259197		0		0		0		0		0
26259198		0		0		0		0		0

	Logout	Downgrade	Save Settings	Submit Downgrade	Cancel	\
0	0	0	0	0	0	
1	0	0	0	0	0	
2	0	0	0	0	0	
3	0	0	0	0	0	
4	0	0	0	0	0	
...	...		...	...		
26259194	0	0	0	0	0	
26259195	0	0	0	0	0	
26259196	0	0	0	0	0	
26259197	0	0	0	0	0	
26259198	0	0	0	0	0	

	Cancellation Confirmation	Register	Submit Registration
0	0	0	0
1	0	0	0
2	0	0	0
3	0	0	0
4	0	0	0
...	...	...	...
26259194	0	0	0
26259195	0	0	0
26259196	0	0	0
26259197	0	0	0
26259198	0	0	0

[26259199 rows x 40 columns]

5 Defining Churn

[7]: *# Define by Button*

```
['NextSong' 'Home' 'Thumbs Up' 'Error' 'Help' 'Add to Playlist'
 'Roll Advert' 'Login' 'Thumbs Down' 'Settings' 'About' 'Add Friend'
 'Upgrade' 'Submit Upgrade' 'Logout' 'Downgrade' 'Save Settings'
 'Submit Downgrade' 'Cancel' 'Cancellation Confirmation' 'Register'
 'Submit Registration']
```